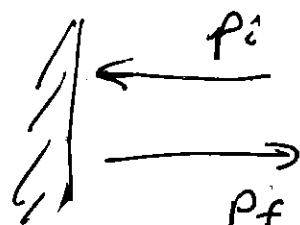
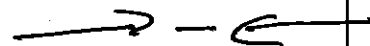


9 Momentum

$$\vec{p} = m \vec{v}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$F = \frac{p_f - p_i}{t}$$



$$F = \frac{p_f - p_i}{t}$$

$$p_i = -p_f$$

$$F = \frac{2p_f}{t}$$

100% rebound

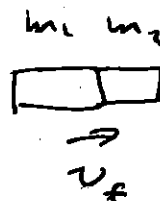
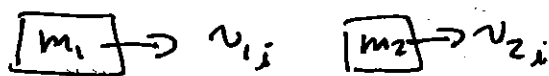
demo

2 types of collision

inelastic p-cons

elastic p-cons & K-cons

inelastic

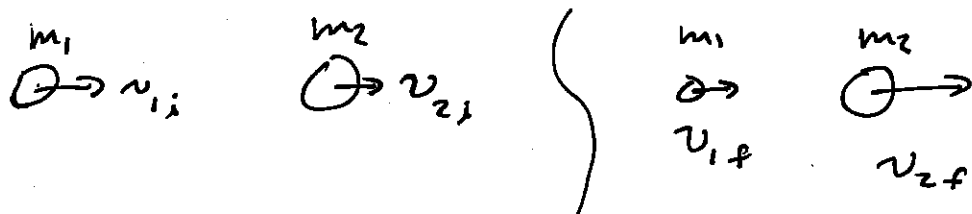


$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

Elastic Collision

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p-cons gives $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$

$$(1) \quad m_1 (v_{1i} - v_{1f}) = m_2 (v_{2f} - v_{2i})$$

K-cons gives $\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$

$$(2) \quad m_1 (v_{1i}^2 - v_{1f}^2) = m_2 (v_{2f}^2 - v_{2i}^2)$$

$$\frac{(2)}{(1)} \quad \frac{m_1 (v_{1i}^2 - v_{1f}^2)}{m_1 (v_{1i} - v_{1f})} = \frac{m_2 (v_{2f}^2 - v_{2i}^2)}{m_2 (v_{2f} - v_{2i})}$$

$$\boxed{v_{1i} + v_{1f} = v_{2i} + v_{2f}}$$

Use to replace Cons of Kinetic Energy

$v_{1f} = ?$ $v_{2f} = v_{1i} + v_{1f} - v_{2i}$ ↓
into $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 (v_{1i} + v_{1f} - v_{2i})$$

$$m_1 v_{1i} - m_2 v_{2i} + 2 m_2 v_{2i} = (m_1 + m_2) v_{1f}$$

3/5

$$v_{1f} = \frac{(m_1 - m_2) v_{1i} + 2 m_2 v_{2i}}{m_1 + m_2}$$

$$8 \quad v_{2f} = v_{1i} + v_{1f} - v_{2i}$$

These go on 3x5 notecard!
Good for any 2 objects hitting elastically

$$\begin{array}{l} \text{---} \rightarrow v_{1i} = 5 \text{ m/s} \quad \text{---} \rightarrow v_{2i} = 2 \text{ m/s} \\ m_1 = 10 \text{ kg} \quad m_2 = 2 \text{ kg} \end{array}$$

$$v_{1f} = ? \quad v_{2f} = ?$$

$$v_{1f} = \frac{(10 - 2) 5 \text{ m/s} + 2 \cdot 2 \cdot 2 \text{ m/s}}{12}$$

$$= \frac{40 + 8}{12} \frac{\text{m}}{\text{s}} = 4 \text{ m/s} \quad v_{1f}$$

$$v_{2f} = 5 + 4 - 2 = 7 \text{ m/s} \quad v_{2f}$$

Example of an elastic collision
Cue shot at 8-ball

4/5



$v_{2i} = 0$

$$v_{1i} = v_{1f} + v_{2f}$$

$$v_{1i} + v_{1f} = v_{2i} + v_{2f}$$

$$= v_{2f}$$

$$v_{1i} = v_{1f} + v_{1i} + v_{1f}$$

$$0 = v_{1f}$$

cue is still

$$v_{1i} = v_{1f} + v_{2f}$$

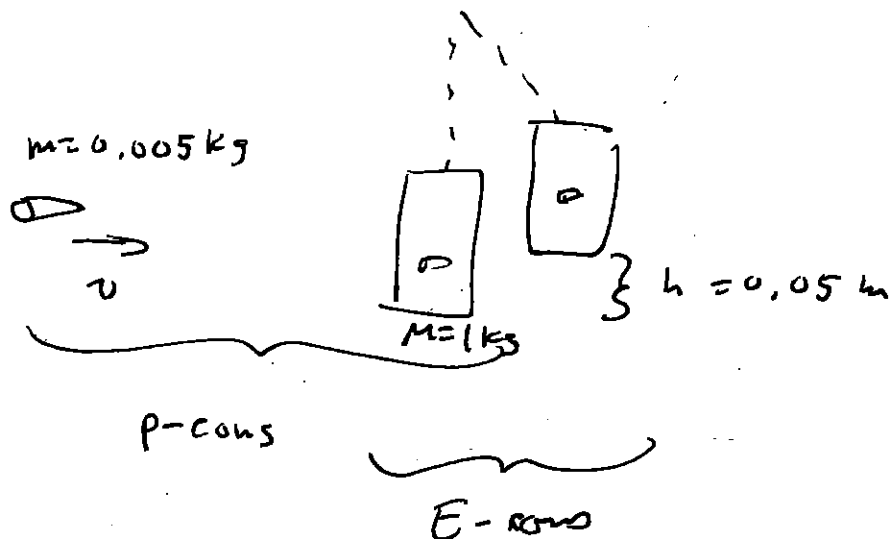
$$v_{1i} = 0 + v_{2f}$$

$$v_{2f} = v_{1i}$$

8 is moving

5/5

Ballistic Pendulum



$$\text{I } mv = (M+m)V$$

$$\text{II } \frac{1}{2} \cancel{(M+m)} V^2 = \cancel{(M+m)} gh$$

$$V = \sqrt{2gh}$$

$$v = \frac{M+m}{m} \sqrt{2gh}$$

$$v = \frac{1.005}{0.005} \sqrt{2 \cdot 9.8 \cdot 0.05}$$

$$= 199 \text{ m/s}$$

$$E_i = E_f + E_{\text{lost}}$$

$$\eta = \frac{E_f}{E_i} = \frac{\cancel{(M+m)} \frac{1}{2} V^2}{\frac{1}{2} m v^2} = \frac{(M+m) gh}{\frac{1}{2} m v^2}$$

$$\eta = \frac{1.005 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 0.05 \text{ m}}{\frac{1}{2} \cdot 0.005 \text{ kg} \cdot (200 \text{ m/s})^2} = \frac{0.5 \text{ J}}{100 \text{ J}} = \frac{1}{2} \%$$